

Madan Bhandari College of Engineering

Urlabari-3, Morang

Final Internal Examination

Level: Bachelor

Full Marks: 100

Programme: B.E(Computer)

Pass Marks: 45

Year/Part: IV/I

Time: 3 hrs

Subject: - Network and Cyber Security(Set-A)

Candidates are required to give their answers in their own words as far as possible.

Attempt all question

1. a) Define computer security. Explain different types of passive and active attacks with appropriate examples. 8
b) Encrypt the plaintext "cyber security is fun" using Caesar cipher using the key equal to the last two digits of your class roll number mod 26. 7
2. a) Explain the encryption and decryption process of DES. Why Double DES failed to provide adequate security. 8
b) Explain the steps involved in RSA algorithm and discuss the basis of its security. 7
3. a) Show that Diffie-Hellman establishes identical keys at both the communicating parties. 7
b) Alice and Bob agree on a prime number $p=23$ and generator $g=5$. 8
Alice chooses private key $a=6$
Bob chooses private key $b=15$
Calculate:
i. Alice's public key
ii. Bob's public key
iii. The shared secret key at both ends
4. a) Define a hash function and explain its properties with examples. Also, explain the concept of collision in hash function. 8^x
b) What is a MIM attack? Discuss the mechanisms used to secure Communication against it. 7
5. a) Discuss common threats in web security and how they are mitigated. 8
b) How is a digital signature verified according to the Digital Signature Standard? 7
6. a) Explain the applications of SSL and describe the SSL/TLS handshake process. 7
b) Explain the OWASP Top 10 Web Application Security Risks. Also discuss why they are important for secure application development? 8
7. Write short notes on: (Any two) 2×5
 - a) Broken access control
 - b) Authentication and Authorization
 - c) Pretty Good Privacy

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- 1.a) What is a network security model? Describe the basic components of a network security model. 8
- b) Using the Play-fair cipher algorithm, encrypt the plain text "CRYPTOGRAPHY" using the keyword "SYMMETRIC". 7
- 2.a) Discuss the need for asymmetric key cryptography. Explain how public key systems overcome the key management and distribution problems of symmetric key cryptography. 8
- b) Explain Advance Encryption standards (AES) for both Encryption and Decryption with figure. 7
- 3.a) What do you mean by Prime Factorization problem? Explain in the context of RSA Algorithm. 7
- b) Consider an ElGamal scheme with a common prime $q=71$ and a primitive root $a=7$
 - i. If B has public key $Y_B=3$ and A chose the random integer $k=2$, what is the cipher text of $M=30$? 8
 - ii. If A now chooses a different value of k so that the encoding of $M=30$ is $C=(59, C_2)$, What is the integer C_2 ? 8
4. a) Explain the architecture of Secure Socket Layer (SSL) with a neat diagram. How TLS improved upon SSL. 8
- b) Explain a scenario where broken access control and insecure session management can lead to data compromise. 7
- 5.a) Explain IPSec architecture and its role in network layer security. 8
- b) Explain various security property of Hash functions. How MAC is implemented? 7
- 6.a) Differentiate with necessary diagrams the IP packet encapsulation in tunneling and transport mode.. 8
- b) Explain OWASP Top 10 web application security risks in detail. 8
7. Write short notes on: (Any two) 2×5
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